

Joan Escamilla

joan.escamilla1@gmail.com · linkedin.com/jescamilla1 · jescamilla1.github.io · (201) 564-8797

Software Engineer II specializing in AI agent and MCP infrastructure engineering, 3+ years building production systems, now building the AI agents and MCP tooling that run them

Experience

RDE Systems | Software Engineer II, AI & Automation

January 2023 - Present

- Maintained full stability for a core EMR module across four separate framework migrations by owning its data-entry screens, reports, and functional modules as subject-matter expert.
- Projected to cut WCAG-compliance audit time by roughly 90 percent (50-60 hours to about 5 hours per system across 50+ systems) by engineering an autonomous agent that scans codebases and generates per-file SVN patches for human review.
- Solved a 40-to-50-hour manual client-import bottleneck, eliminating manual SSMS/SVN lookups across two additional dev-workflow tools and scoping team-wide hosting within existing AWS budget, by architecting the company's first MCP server infrastructure and leading a cross-functional meeting with AWS engineers on AgentCore, Bedrock, and Lambda.
- Cut routine SQL scripting time roughly in half, and complex schema-design work by a similar margin, by shipping an agent that automates the team's unwritten SQL conventions.
- Delivered a fully modernized frontend in 3 days, replacing 20-year-old default Bootstrap styling on the company's internal site and now extended into a formal design system led with 2 junior developers on a 2-week timeline, by directing an AI-guided CSS rebuild of a 30-year-old product UI.
- Converted a university capstone program into a hiring pipeline, selecting a 5-developer team and hiring one into a junior role now a year into the job, by leading the program and continuing direct mentorship post-hire.
- Reduced ramp-up time for new developers across four new hires joining the framework by overhauling the RDE onboarding program.

The Coder School | Programming Tutor

September 2021 - July 2023

- Designed and taught a project-based curriculum in React JS, Python, and Java, giving students hands-on experience completing multiple real projects per month.
- Tracked progress across 20+ students with regular parent feedback; students averaged a 30% improvement in programming skills by internal metrics.

MetroStar | Frontend Developer Intern

June 2022 - August 2022

- Designed and built interactive Tableau dashboards using U.S. Census data, giving city planners and officials in Bloomington, Indiana a faster way to identify economic and urban development opportunities.
- Converted Tableau's JS API into reusable React components, cutting dashboard integration time for new use cases across the team.

Education

New Jersey Institute of Technology | B.S Computer Science

May 2023

Certifications

AI Fluency: Framework & Foundations | Anthropic

April 2026

Teaching AI Fluency | Anthropic

April 2026

Introduction to Model Context Protocol | Anthropic

July 2026

Skills

Programming Languages: ColdFusion, SQL, PHP, Python, JavaScript, CSS, HTML

AI/Agentic Development: Agentic AI, MCP Server Development, Workflow Automation, Prompt Engineering, Anthropic Claude, GitHub Copilot

Tools/Frameworks: React JS, jQuery, SQL Server, SVN, Tableau, Figma, Tailwind CSS, Bootstrap, Git, AWS (AgentCore, Bedrock, Lambda)